

Lecture 6

Diagonalization, Cayley–Hamilton, Functions of Operators

Throughout, $T \in \mathcal{L}(V)$ and $\chi_T(\lambda) = \det(\lambda I - T)$ is its characteristic polynomial. Tags indicate the flavour of each problem; a ★ marks a harder one.

Core tutorial route

Work **6.1(i), 6.1(iv), 6.2(ii), 6.3, 6.5, 6.6(i), 6.7(ii), 6.10(ii), and 6.10(iv)** in that order. This route covers diagonalization and matrix powers, a common eigenbasis, the Cayley–Hamilton theorem, spectral mapping, and linear dynamics.

Additional practice: 6.1(ii)–(iii), 6.1(v), 6.2(i), 6.4, 6.6(ii), 6.7(i), 6.8(i)–(ii), 6.10(i). **Bridge:** 6.9(ii), 6.10(iii) (convergence and matrix dynamics). **Extension:** 6.8(iii), 6.9(i), 6.11–6.14 (the determinant–trace identity, projections, the lemma for the general proof of Cayley–Hamilton, generalized eigenvectors, the Jordan–Chevalley decomposition, the exponential of a defective matrix, ladder operators).

Diagonalization and commuting operators

Exercise 6.1 (computational) Decide whether the matrix A is diagonalizable and, if it is, find matrices P and D such that $P^{-1}AP = D$, where (i) $A = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$; (ii) $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ (over $\mathbb{R}$); (iii) $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ (over $\mathbb{C}$); (iv) $A = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$; (v) $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

Answer. (i) Eigenvalues 1, 4; $P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \text{diag}(4, 1)$. (ii) Eigenvalues 3, -1 ; $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \text{diag}(3, -1)$. (iii) Eigenvalues $2 \pm i$; $P = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$, $D = \text{diag}(2 + i, 2 - i)$. (iv) Not diagonalizable: the only eigenvalue 5 has a one-dimensional eigenspace. (v) Not diagonalizable: the only eigenvalue 2 has a one-dimensional eigenspace.

Exercise 6.2 (computational) Compute A^n for every $n \geq 0$, where (i) $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (its eigenvalues are 3 and 1, with eigenvectors $(1, 1)$ and $(1, -1)$); (ii) $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

Answer. (i) $A^n = \frac{1}{2} \begin{pmatrix} 3^{n+1} & 3^n - 1 \\ 3^n - 1 & 3^{n+1} \end{pmatrix}$. (ii) $A^n = \frac{1}{2} \begin{pmatrix} 3^n + (-1)^n & 3^n - (-1)^n \\ 3^n - (-1)^n & 3^n + (-1)^n \end{pmatrix}$.

Exercise 6.3 (computational) Let $S = \text{diag}(1, 1, 2)$ and $T = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$. (i) Check that $ST = TS$. (ii) The standard basis consists of eigenvectors of S . Does it consist of eigenvectors of T ? (iii) Find a basis of $\mathbb{R}^3$ in which the matrices of both S and T are diagonal.

Answer. (i) $ST = TS = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}$. (ii) No: $Te_1 = (2, 1, 0)$. (iii) $(1, 1, 0)$, $(1, -1, 0)$, $(0, 0, 1)$; in it $S = \text{diag}(1, 1, 2)$ and $T = \text{diag}(3, 1, 4)$.

Hint. (iii) The operator T maps the eigenspace $E(1, S) = \text{span}(e_1, e_2)$ into itself. Look for the eigenvectors of T inside it.

The Cayley–Hamilton theorem

Exercise 6.4 (computational) Use the Cayley–Hamilton theorem to express the matrix A^{-1} as a polynomial in A , and compute it, where (i) $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$; (ii) $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$; (iii) $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix}$ (compute the characteristic polynomial χ_A first).

Answer. (i) $A^{-1} = \frac{1}{2}(A - 5I) = \begin{pmatrix} \frac{-2}{2} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$. (ii) $A^{-1} = \frac{1}{10}(7I - A) = \begin{pmatrix} \frac{2}{5} & -\frac{1}{10} \\ -\frac{1}{5} & \frac{3}{10} \end{pmatrix}$. (iii) $\chi_A(\lambda) = \lambda^3 - 7\lambda^2 + 16\lambda - 12$; $A^{-1} = \frac{1}{12}(A^2 - 7A + 16I) = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ -\frac{1}{6} & \frac{1}{3} & 0 \\ \frac{1}{12} & -\frac{1}{6} & \frac{1}{2} \end{pmatrix}$.

Exercise 6.5 (computational) Let $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$. Use the Cayley–Hamilton theorem to write A^4 in the form $\alpha A + \beta I$, and compute the resulting matrix.

Answer. $A^4 = 15A - 14I = \begin{pmatrix} -14 & 15 \\ -30 & 31 \end{pmatrix}$.

Exercise 6.6 (proof)

- (i) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Show by direct computation that $A^2 - (\text{tr } A)A + (\det A)I = 0$. This verifies the Cayley–Hamilton theorem for every 2×2 matrix.
- (ii) Suppose $T \in \mathcal{L}(V)$ is an invertible operator and $\dim V = n$. Prove that T^{-1} is a polynomial in T .

Hint. (i) Compute A^2 and $(\text{tr } A)A$ and subtract; the off-diagonal entries cancel, and what is left is a multiple of I . (ii) Write out the Cayley–Hamilton identity for T and isolate the constant term $c_0 I$. It is invertibility that makes $c_0 = (-1)^n \det T$ nonzero, so you may divide by it.

Functions of operators and linear dynamics

Exercise 6.7 (computational)

- (i) Find the square root $\sqrt{A}$ of the matrix $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ that has positive eigenvalues and satisfies $(\sqrt{A})^2 = A$.
- (ii) Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with $\text{spec}(A) = \{1, 3\}$. Use spectral mapping to find $\text{spec}(B)$ for $B = A^3 - 2A + I$.

Answer. (i) $\sqrt{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. (ii) $\text{spec}(B) = \{0, 22\}$ (with the same eigenvectors $(1, -1)$ and $(1, 1)$).

Exercise 6.8 (proof)

- (i) Suppose $T \in \mathcal{L}(V)$ is diagonalizable and has only one eigenvalue λ . Prove that $T = \lambda I$.
- (ii) Suppose T is diagonalizable and $T = PDP^{-1}$. Prove that for any polynomials p, q the operators $p(T)$ and $q(T)$ commute.
- (iii) Let A be a diagonalizable matrix. Prove that $\det(e^A) = e^{\text{tr } A}$.

Hint. (i) Diagonalize: $T = PDP^{-1}$, where every diagonal entry of D is the same λ , and see what that leaves. (ii) Move everything to D : $p(T) = P p(D) P^{-1}$, and diagonal matrices commute. Note also that diagonalizability is not actually needed — it is enough that the powers of T commute. (iii) A similarity transformation changes neither the determinant nor the trace, and $e^{PDP^{-1}} = P e^D P^{-1}$, where e^D is a diagonal matrix. What remains is $\prod_k e^{\lambda_k}$.

Exercise 6.9 (★)

- (i) Suppose T is diagonalizable and its eigenvalues lie in the set $\{0, 1\}$. Prove that $T^2 = T$.
- (ii) Suppose A is a diagonalizable matrix and each of its eigenvalues satisfies $|\lambda| < 1$. Prove that $A^n \rightarrow 0$ as $n \rightarrow \infty$ (componentwise).

Hint. (i) Diagonalize T ; the claim reduces to checking $\lambda^2 = \lambda$ for each eigenvalue. (ii) Diagonalize and examine D^n componentwise; the matrices P and P^{-1} do not depend on n .

Exercise 6.10 (computational)

- (i) Compute e^{tA} , where $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.
- (ii) Compute e^{tA} , where $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.
- (iii) Solve the equation $\dot{x} = Ax$ with $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $x(0) = (2, 0)$.
- (iv) Let $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$. Compute e^{tA} and describe the trajectories of the solutions of $\dot{x} = Ax$.

Answer. (i) $e^{tA} = \frac{1}{2} \begin{pmatrix} e^{3t} + e^t & e^{3t} - e^t \\ e^{3t} - e^t & e^{3t} + e^t \end{pmatrix}$. (ii) $e^{tA} = \frac{1}{2} \begin{pmatrix} e^{3t} + e^{-t} & e^{3t} - e^{-t} \\ e^{3t} - e^{-t} & e^{3t} + e^{-t} \end{pmatrix}$. (iii) $x(t) = (e^{3t} + e^{-t}, e^{3t} - e^{-t})$. (iv) $e^{tA} = \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix}$; the trajectories are circles traversed at angular frequency 2.

Triangular matrices and generalized eigenvectors [self-study]

Exercise 6.11 (proof) Suppose that, with respect to the basis $v_1, \dots, v_n$, the matrix of T is upper triangular with diagonal entries $\lambda_1, \dots, \lambda_n$. Prove that $(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$.

Hint. Triangularity gives $(T - \lambda_k I)v_k \in \text{span}(v_1, \dots, v_{k-1})$. Use induction on k and the fact that the factors commute, so they can be applied in any order.

Exercise 6.12 (computational) Find the eigenspace $E(3, A)$ and the generalized eigenspace $G(3, A)$ of the matrix $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$. Write down the Jordan chain.

Answer. $E(3, A) = \text{span}((1, 0))$ (dim 1), $G(3, A) = \mathbb{C}^2$ (dim 2); Jordan chain $(0, 1) \rightarrow (1, 0) \rightarrow 0$.

Exercise 6.13 (computational)

- (i) Write the matrix $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ in the form $A = D + N$, where D is diagonalizable, N is nilpotent and $DN = ND$. Use this decomposition to compute e^{tA} .
- (ii) Compute e^{tA} for $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

Answer. (i) $D = 2I$, $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$; $e^{tA} = e^{2t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. (ii) $e^{tA} = \begin{pmatrix} e^{2t} & te^{2t} & 0 \\ 0 & e^{2t} & 0 \\ 0 & 0 & e^{3t} \end{pmatrix}$.

Exercise 6.14 (★) On the state space of spin 1, the raising operator acts by $J_+|-1\rangle = \sqrt{2}|0\rangle$, $J_+|0\rangle = \sqrt{2}|1\rangle$, $J_+|1\rangle = 0$. Write its matrix in the basis $(|1\rangle, |0\rangle, |-1\rangle)$ and show that J_+ is nilpotent.

Hint. The columns of the matrix are the images of the basis kets. A strictly upper triangular matrix is nilpotent — compute its powers.